Cheatsheet AZ-104 — 09 · VM Scale Sets, autoscale, extensions and VM access

1. VMSS: Flexible vs Uniform

	Flexible (recommended, default)	**Uniform**
VM type	**Standard** VMs (Microsoft.Compute/virtualMachines): RBAC, tags, Backup, Site Recovery, normal API	Child VMs of the scale set (own API); **no Backup or ASR**
Maximum instances	**1,000**	100 with FD guarantee (up to 3,000 with multiple placement groups)
Mixing	Sizes, **Linux and Windows**, **Spot + regular** in the same set; add existing VMs	All identical (same image, size and priority)
Zones and fault domains	Spreads across 1-3 zones; max FD 3 (1 if zonal); **no update domains**; zone / FD can be pinned per VM	Zones 1-3; FD 2, 3 or 5; **5 update domains**
Scaling and health	Autoscale, scale-in policy, instance protection, instance repair (with **Application Health extension**), upgrade policy Automatic / Rolling / Manual	Same; plus **automatic OS image upgrades** (still preview on Flexible) and health via **LB health probe**; **overprovisioning**
Networking	**Standard Load Balancer** / App Gateway only; **no outbound internet by default** (NAT Gateway, LB outbound rule or PIP)	Supported Basic LB (retired in Sept. 2025) and NAT pool
Identity	**User-assigned** managed identity only	System or user-assigned
SLA	99.95 % in FDs · **99.99 %** across several zones	Same

- The mode is chosen **at creation** and cannot be changed. "Mixed VM sizes", "Spot and regular", "attach an existing VM" → Flexible. "Service Fabric / AKS node pools", "automatic OS upgrades", "overprovisioning" → Uniform.
- New instances come from the model's **image** (Marketplace or **Compute Gallery**); initial configuration is done with **extensions** (Custom Script), cloud-init or a pre-built image.

2. Autoscale (VMSS and App Service)

Element	**What it is**	**Detail**
Profile (scale condition)	Up to **20** per resource: **Default** (one), **Recurrence** (days and hours), **Fixed date** (specific event)	Evaluation order: fixed date → recurrence → default
Capacity	**Minimum / Maximum / Default**	Default = what it falls back to if it **cannot read the metric** (never scales below it in that case)
Rule	Up to **10** per profile; metric (Percentage CPU, Available Memory Bytes, Network In/Out, Disk, Storage / Service Bus **queue length**, custom metrics), statistic, **duration** (window, e.g. 10 min), **time grain** (1 min), operator and threshold, action (**increase / decrease** count, percent or exact), **cooldown** (default **5 min**, per rule)	One scale-out rule and one scale-in rule; "> 75 % for 10 min → +2; < 25 % → −1"
Evaluation	Every 30-60 s; if several scale-out rules fire, the **largest** capacity wins; it only **scales in** if **all** scale-in rules are met; avoids **flapping** (does not scale in if that would trigger a scale-out)	Cooldown is counted per rule from the last change
Predictive autoscale	Scales ahead of time with ML on CPU (VMSS only)	"scale before the daily peak"
Scale-in policy	**Default** (balances zones, then FDs, deletes the highest ID), **NewestVM**, **OldestVM**	"remove the oldest instances first"
Instance protection	Protect an instance from scale-in or from all set operations	Keep a diagnostics VM
Notifications	Email to admins / co-admins / addresses, **webhook**	—

- Manual: set the instance count (`az vmss scale --new-capacity 5`). Scheduled: recurrence profile (business hours 10 instances, night 2). External metric: Storage queue → `dividePerInstance`.
- In **App Service** autoscale belongs to the **plan** (affects all its apps) and needs Standard or higher; Premium v2/v3 also has managed **automatic scaling** (cheatsheet 11).

3. Updating the model and keeping instances healthy

Piece	**What it does**
Upgrade policy	**Manual**: you change the model and apply it yourself instance by instance (`az vmss update-instances --instance-ids "*"`) · **Automatic**: all at once, no order (downtime) · **Rolling**: in batches (max %, max unhealthy %, pause) and requires a **health probe** (Uniform) or the **Application Health extension**
Automatic OS image upgrades	Uniform: new version of the Marketplace / gallery image → automatic rolling upgrade; requires health configured
Automatic VM guest patching	Flexible (and standalone VMs): patches inside the OS with platform orchestration
Instance repair	Replaces unhealthy instances (grace period); needs the Application Health extension or a probe
Reimage / Upgrade	Reimage reinstalls the instance from the image; Upgrade applies the **latest model** ("Latest model: No" = pending)
Terminate notification	Notice (up to 15 min) before deleting an instance so it can be drained

4. VM extensions

Extension	Purpose	Exam details
Custom Script Extension (Microsoft.Compute / CustomScriptExtension; Linux: Microsoft.Azure.Extensions / CustomScript)	Download and run a script **after** deployment (install IIS, configure)	`fileUris` + `commandToExecute`; downloads from Storage (key, SAS or **managed identity**) or GitHub; **90 min** maximum; runs **once** (to rerun: change `timestamp` or `forceUpdateTag`); **only one** per VM; secrets in `protectedSettings`; logs in `C:\WindowsAzure\Logs\Plugins\...` or `/var/log/azure/`; do not reboot inside the script; runs as SYSTEM / root
cloud-init (custom data)	Configuration on the **first boot** of Linux (packages, users, files)	Linux only, passed with `--custom-data`, runs **before** the extensions; **user data** (both OSes) is queried from IMDS but is not secret
Azure Monitor Agent (AMA)	Collect OS metrics and logs for Log Analytics / Monitor with **Data Collection Rules**	Replaces the Log Analytics agent (MMA, retired Aug. 2024); needs managed identity; VM insights = AMA + **Dependency agent**
Network Watcher agent	Packet capture, connection troubleshoot, Connection monitor	Installed automatically when using those tools
VMAccess	Reset password, user, SSH keys, RDP	Help → Reset password; `az vm user update`
AADLoginForWindows / AADSSHLoginForLinux	Sign in with **Entra ID**	Plus the Virtual Machine Administrator Login / User Login roles; MFA via Conditional Access
Microsoft Antimalware	Antivirus on Windows	Defender for Endpoint via Defender for Cloud is the modern option
Key Vault VM extension	Pull certificates from Key Vault and renew them	—
DSC / Machine Configuration	Desired state (DSC being retired → **Azure Machine Configuration** with Azure Policy)	"audit a setting inside the VM" → guest configuration policy
Azure Disk Encryption	BitLocker / dm-crypt (cheatsheet 08)	—

- States: Provisioning succeeded / **transitioning** (in progress) / **failed** (check the logs; cause no. 1: script error or URL unreachable because of an NSG / storage firewall). Extensions need the **VM Agent** (included in Marketplace images). In VMSS, better to deploy the extension with CLI / PowerShell / template and managed identity: the portal's SAS expires and breaks scale-out.

5. Automation around VMs

Tool	Purpose	Signal
Run command	One-off script without network or RDP (`az vm run-command invoke`)	"cannot RDP, fix the firewall"
Azure Automation	PowerShell / Python runbooks with **schedules**, webhooks, **Hybrid Runbook Worker** for on-prem; State Configuration (being retired)	"run a script every night", "stop VMs at 19:00"
Start/Stop VMs v2	Functions-based solution to start / stop by schedule or CPU	"start VMs at 8:00 and stop at 19:00"
Auto-shutdown	Only shuts down at a set time, with notification	"shut down dev VMs daily"
Azure Update Manager	Patching, maintenance schedules, patch orchestration	"install updates on a schedule"
Azure Policy (guest configuration)	Audit / enforce configuration inside the OS	"audit VMs where a service is disabled"
VM Image Builder	Build custom images in a pipeline	"build images automatically"

6. Accessing VMs

Method	How	Detail
RDP 3389 / SSH 22 over public IP	NSG with a rule from specific IPs; **Just-in-time** (Defender for Cloud) opens the port only on request and for a limited time	Avoid "Any" as the source
Azure Bastion	RDP / SSH **in the browser** (port 443) with no public IP on the VM; **AzureBastionSubnet /26 or larger** subnet, **Standard static** public IP; reaches VMs in **peered VNets**	SKUs: **Developer** (free, no subnet, limited) · **Basic** (2 instances) · **Standard** (host scaling 2-50, **native client** mstsc / ssh, **file transfer**, **shareable links**, custom ports, connection by IP) · **Premium** (+ **session recording**, private-only)
Serial console	Console without network	Boot diagnostics enabled
Entra ID login	Extension + role; MFA with Conditional Access	"no local admin accounts", "require MFA to RDP"
VPN / ExpressRoute	RDP to the **private** IP from on-prem	Cheatsheet 13
Run command / VMAccess	Repair without logging in	—

7. Commands that show up

Task	PowerShell	Azure CLI
Create VMSS	`New-AzVmss -ResourceGroupName RG1 -VMScaleSetName vmss1 -Location westeurope -OrchestrationMode Flexible -ImageName Ubuntu2204 -InstanceCount 2 -Zone 1,2,3 -UpgradePolicyMode Rolling`	`az vmss create -g RG1 -n vmss1 --image Ubuntu2204 --orchestration-mode Flexible --instance-count 2 --zones 1 2 3 --vm-sku Standard_D2s_v5 --upgrade-policy-mode Rolling --lb lb1 --admin-username azureuser --generate-ssh-keys`
Scale manually	`Update-AzVmss -ResourceGroupName RG1 -VMScaleSetName vmss1 -SkuCapacity 5`	`az vmss scale -g RG1 -n vmss1 --new-capacity 5`
Autoscale	`$r = New-AzAutoscaleScaleRuleObject -MetricTriggerMetricName "Percentage CPU" -MetricTriggerMetricResourceUri <vmss id> -MetricTriggerOperator GreaterThan -MetricTriggerStatistic Average -MetricTriggerThreshold 75 -MetricTriggerTimeAggregation Average -MetricTriggerTimeGrain (New-TimeSpan -Minutes 1) -MetricTriggerTimeWindow (New-TimeSpan -Minutes 10) -ScaleActionDirection Increase -ScaleActionType ChangeCount -ScaleActionValue 1 -ScaleActionCooldown (New-TimeSpan -Minutes 5) → $p = New-AzAutoscaleProfileObject -Name p1 -CapacityDefault 2 -CapacityMinimum 2 -CapacityMaximum 10 -Rule $r → New-AzAutoscaleSetting -Name as1 -ResourceGroupName RG1 -Location westeurope -TargetResourceId <vmss id> -Profile $p`	`az monitor autoscale create -g RG1 --resource vmss1 --resource-type Microsoft.Compute/virtualMachineScaleSets --min-count 2 --max-count 10 --count 2 · az monitor autoscale rule create -g RG1 --autoscale-name vmss1 --condition "Percentage CPU > 75 avg 10m" --scale out 2 · --condition "Percentage CPU < 25 avg 10m" --scale in 1`
Apply model	`Update-AzVmssInstance -ResourceGroupName RG1 -VMScaleSetName vmss1 -InstanceId "*"`	`az vmss update-instances -g RG1 -n vmss1 --instance-ids "*" · az vmss update -g RG1 -n vmss1 --set upgradePolicy.mode=Rolling`
Custom Script	`Set-AzVMCustomScriptExtension -ResourceGroupName RG1 -VMName vm1 -Location westeurope -FileUri https://.../install.ps1 -Run install.ps1 -Name cse · VMSS: Add-AzVmssExtension`	`az vm extension set -g RG1 --vm-name vm1 --name CustomScriptExtension --publisher Microsoft.Compute --settings '{"fileUris":["https://.../install.ps1"],"commandToExecute":"powershell -ExecutionPolicy Unrestricted -File install.ps1"}' · Linux: --name CustomScript --publisher Microsoft.Azure.Extensions · VMSS: az vmss extension set`
List extensions	`Get-AzVMExtension -ResourceGroupName RG1 -VMName vm1`	`az vm extension list -g RG1 --vm-name vm1 -o table · az vm get-instance-view`
Entra login	`Set-AzVMExtension -Publisher Microsoft.Azure.ActiveDirectory -ExtensionType AADLoginForWindows -Name AADLogin ... + New-AzRoleAssignment -RoleDefinitionName "Virtual Machine Administrator Login"`	`az vm extension set --publisher Microsoft.Azure.ActiveDirectory --name AADSSHLoginForLinux ... + az role assignment create --role "Virtual Machine User Login" --assignee ... --scope <vm id>`
Bastion	`New-AzBastion -ResourceGroupName RG1 -Name bas1 -PublicIpAddressRgName RG1 -PublicIpAddressName pip-bas -VirtualNetworkRgName RG1 -VirtualNetworkName vnet1 -Sku Standard`	`az network bastion create -g RG1 -n bas1 --public-ip-address pip-bas --vnet-name vnet1 --sku Standard -l westeurope · native client: az network bastion rdp --name bas1 -g RG1 --target-resource-id <vm id>`
Run command	`Invoke-AzVMRunCommand -ResourceGroupName RG1 -VMName vm1 -CommandId RunPowerShellScript -ScriptString "Set-NetFirewallRule ..."`	`az vm run-command invoke -g RG1 -n vm1 --command-id RunPowerShellScript --scripts "Get-Service"`

8. Decision tree

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What does the wording ask for?
├─ Scale identical VMs automatically ..... VMSS (Flexible) + autoscale
├─ Mix sizes, Spot and regular, or add existing VMs ..... VMSS Flexible
├─ Automatic OS image upgrades or overprovisioning ..... VMSS Uniform
├─ Scale by schedule (office hours / night) ..... Autoscale profile Recurrence
├─ Scale by message queue ..... Rule with Storage Queue / Service Bus metric
├─ Prevent a specific instance from being deleted ..... Instance protection
├─ Update without downtime ..... Upgrade policy Rolling + Application Health extension
├─ Install software after deployment ..... Custom Script Extension (or cloud-init on Linux)
├─ Run the script again ..... Change timestamp / forceUpdateTag
├─ Collect OS logs into Log Analytics ..... Azure Monitor Agent + Data Collection Rule
├─ Reset the admin password ..... VMAccess (Help → Reset password)
├─ RDP without a public IP ..... Azure Bastion (native client → Standard)
├─ Record RDP sessions ..... Bastion Premium
├─ Sign in with Entra ID and MFA ..... AADLogin extension + VM Admin/User Login role + CA
├─ Shut down dev VMs every night ..... Auto-shutdown (or Start/Stop v2 if they must start too)
└─ Repair the VM without network ..... Run command / Serial console
```

9. Signals → answer

Wording	Answer
"scale out when CPU exceeds 70 % for 10 minutes"	Autoscale rule: Percentage CPU > 70, duration 10 min, increase
"instances keep scaling in and out"	Flapping: separate the thresholds, raise the cooldown
"autoscale uses default count"	It could not read the metric (agent / diagnostics)
"more instances on weekdays 8-18"	Recurrence profile
"mix Spot and regular instances"	Flexible
"scale set with Basic Load Balancer"	Uniform only (and Basic LB is retired: use Standard)

Wording	Answer
"new instances don't have the app installed"	Custom Script Extension in the VMSS model (or gallery image)
"script ran once, needs to run again"	timestamp / forceUpdateTag
"extension failed: cannot download the script"	NSG / storage firewall is blocking; use managed identity or open it
"script takes longer than 90 minutes"	Fails: split it or use Automation
"configure Linux VM at first boot"	cloud-init (custom data)
"collect Windows event logs to Log Analytics"	AMA + DCR
"Log Analytics agent", "MMA"	Retired: migrate to AMA
"RDP to VMs without public IPs from the portal"	Bastion
"Bastion subnet /27 fails"	AzureBastionSubnet minimum **/26**
"connect with mstsc through Bastion"	Standard SKU (native client)
"copy files through Bastion"	Standard (file transfer)
"VM in a peered VNet through Bastion"	Supported: one Bastion in the hub
"update the scale set model but instances unchanged"	Upgrade policy Manual: `update-instances`
"prevent the old instance from being removed"	Instance protection

10. Mistakes to avoid

1. Choosing Uniform when mixed sizes or Spot are required, or Flexible when automatic OS upgrades are required.
2. Setting a single autoscale rule (without the scale-in one) or overlapping thresholds (flapping).
3. Expecting a Custom Script Extension to run on every boot or twice without changing the timestamp.
4. Putting secrets in `settings` instead of `protectedSettings`.
5. Deploying the extension on a VMSS with a portal SAS that expires.
6. Creating AzureBastionSubnet with /27 or with another name, or expecting the native client on Basic.
7. Using the Log Analytics agent (MMA) instead of AMA.
8. Assuming auto-shutdown also starts the VMs.